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9428–30 DISCUSSION TRANSPORTATION ENGINEERING GROUP

Singapore Mass Rapid Transit System

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Authors' introduction

In this introduction, we should like to elaborate on some financial and ridership aspects. The programme for construction was originally determined by the decision to limit expenditure on the MRT to an average of 5% of that part of the annual budget allocated to development. During the construction, the limit could be breached at peak periods but, overall, the average was to be 5%. The financial planning was based necessarily on assumptions of the annual growth rate of the gross national product, and two scenarios of 8% and 10% annual increase were considered. Table 1 shows the calculation at 8% based on a 1981 start and a 12 year programme. The annual growth rate actually achieved exceeded the forecast 8%, and the delay in the start of construction to 1984 meant that, in practice, rather larger resources were available to the Corporation. This effectively determined the original nine-year construction programme.

Table 1. MRT investment rate as a percentage of development budget

Year	Development budget: S$ million	5% of budget	Annual expenditure (proposed construction programme)
1979–80	2560	128	
1980–81	2765	138	
1981–82	2986	149	75
1982–83	3225	161	122
1983–84	3483	174	326
1984–85	3761	188	521
1985–86	4062	203	600
1986–87	4387	219	641
1987–88	4738	237	537
1988–89	5117	256	375
1989–90	5527	276	257
1990–91	5969	298	187
1991–92	6447	322	166
1992–93	6963	348	94
1993–94	7520	376	45
Total (1982–94)		3058	3946
Assumed annual growth rate in real terms = 8%			

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Table 2. Contract award with financing

Contract no.	Currency	Finance offer
101	US$	Deferred payment of 70% of contract price
107B	US$	Deferred payment of 70% of contract price
203	US$	Deferred payment of 100% of contract price
204	US$	Deferred payment of 95% of contract price
301	US$	Deferred payment of 70% of contract price
304	S$ and US$	Deferred payment of US$ portion
404	US$	Deferred payment of 95% of contract price
405	S$ and US$	Deferred payment of US$ portion

Table 3. Passenger numbers

Section opened	No. of station	Opening date	Number of passengers
Yio Chu Kang—Toa Payoh	5	07.11.87	40 000
Toa Payoh—Outram Park	9	12.12.87	155 000
Outram Park—Clementi	5	12.03.88	220 000
Clementi—Lakeside	3	05.11.88	270 000
Yio Chu Kang—Yishun	3	20.12.88	290 000

2. Cash flow, currency and financial package could all assist in reducing the net value of contractors' bids. The Corporation indicated its preferred currencies to contractors during the tender period, and bids in US dollars, French francs and pounds sterling were received. Financing offers were accepted in only eight cases (Table 2) out of a total of 39 main contracts, because the offers were generally not sufficiently attractive to Singapore whose own resources are very advantageously invested. In one case, the contractor offering a financial package was the lowest anyway.

3. Overall, the majority of claims and variations have now been settled, and the project is running at approximately 7·1/2% over the sum of the tendered prices. It is estimated that the out-turn cost will be some 15% less than the original budget of S$5000 million at 1982 prices.

4. Passenger traffic has been approximately in accordance with expectations at the progressive opening stages (Table 3). On completion of the system as presently authorized, a daily ridership approaching 600 000 passengers is anticipated.

A. J. Howard, *Mott Macdonald*

When reading Paper 9428, I found it interesting to compare and contrast the Singapore MRT with the early development of the London Underground.

6. The City and South London Railway became the Northern line, which was gradually extended to Stockwell, and eventually to Morden. The philosophy was that if the railway was put there, people would go and live by it, so that they could travel in to the centre of London. It was a very imaginative and rather brave idea, to which there was a lot of opposition; but it worked, as is now known.

7. Provision for the future railway in new buildings was mentioned, and I know one or two holes in the basements of buildings in London which are awaiting railways which will probably never be constructed. In the UK, there has not been the same ability to plan, and the planners do not always get it right. For

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example, a decision was made to develop the London Docklands, whereupon the Docklands Light Railway was built. However, before it was opened, the Canary Wharf development was initiated and turned everything on its head. The light railway was found to be inadequate; it has had to be strengthened; and it had to be extended into the City of London in what must be some of the most difficult circumstances. At the same time, there is the new situation in which private developers are prepared to contribute considerable sums of money for construction of a public underground railway.

8. In contrast, Singapore seems to have been planning this railway for a very long time. I had the impression that the new towns are being planned with the railway as a part of the planning. The land area is, of course, much smaller and the circumstances are different, but the Singapore Government is obviously very committed to the overall planning of the island. I wonder whether the MRT has had an effect on traffic congestion in the middle of town? Are people moving to outer areas, using the rapid transit system, and has it replaced traffic in the middle of the town; or is the traffic situation as bad as ever because people still want to use their cars? Would buses have coped with what has happened in Singapore? A bus system was obviously the leading contender at one time following the Harvard Report.

9. It was a very bold decision, I think, to accelerate construction to take advantage of what was seen as a downturn in the world market, particularly in tunnelling. Of course, in the out-turn some very favourable contract prices were obtained, but did the contractors appreciate this in the long term? Sometimes when very tight tenders are obtained, there is imaginative financial juggling, and claims at the end of the job, particularly when a great deal of risk is borne by the contractor. I know Clause 12 was used, but the contractors had the exchange rate fluctuation risk, and also had the risk of major and minor government legislation. How could the contractor have been expected to foresee what the government might do?

A. C. D. Malcolm *Crown Agents*

Long before the MRT was ever designed, the Crown Agents were involved with the project. Their report showed that the causeway (which was itself built by the Crown Agents) should carry the top end of the metro into Malaysia, or, as it was then, Malaya, and this has not been done. Could the Authors say why this was not done?

11. The Authors, quite rightly, mention the funds in Singapore; but where did they come from? Was it not by good management of the colony funds in Singapore for over one-hundred years?

12. It was very interesting to see how the design was carried out, but why was the line not taken up the Bukit Timah Road? It is true that Mile 11 is huge in asbestos cement manufacture and that Mile 31 is somebody else again, but did the Authors not think that there was potential there?

13. Could the Authors say more about how the MRT was constructed—cut and cover, boring, etc.? It would be interesting to know about the equipment used.

I. H. Elliott, *Mott MacDonald*

I should like, in particular, to draw attention to the phase III study, because I believe that the way the Singapore government chose to carry out this exercise at that stage has been a very major contribution to the success of the project. My

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involvement in this has been very small but the Authors all made a major contribution. The work was carried out by a single multidiscipline team of public and private sector consultants from the UK, together with the local Singapore government staff led by Mr Hulme.

15. Essentially, the objectives were fairly simple: to establish a route, to establish the cost, and to plan ahead for the design. This obviously involved an integrated and interactive study with the nine main disciplines, namely railway operation, rolling stock, mechanical equipment, signalling and communications, electrical services and equipment, ventilation and cooling system, architecture, geotechnical engineering, and civil and structural engineering. Paragraph 11 of Paper 9428 records that the essential recommendations of phase III have been implemented, and elsewhere it is recorded that the cost estimates have been proved to be reliable. I agree with the Authors that the use of a resource-based costing procedure is one of the reasons for that. Could the Authors comment on how important they feel this phase of the project now is.

16. I did identify three items which came out of Phase III, which I felt to be of interest. The first is the idea of having the intersection between the east–west and the north–south lines, essentially at two stations rather than at one station as was originally planned in phase II. I should be interested in comments on the practicalities of the construction, and on how effective it is operationally. The second feature is the use of grade separation for the approach to the depot, which has been illustrated in Paper 9429. The third item is in relation to considerations of passenger comfort, in view of Singapore's year-round high temperatures and high humidity, and of the fact that passenger comfort was of major importance. Three systems of air conditioning and ventilation were considered; but, although in the planning of phase III, a conventional open system had been recommended, it was suggested that, with the very significant savings that were potentially available with a segregated system, the latter should be further developed. As is mentioned in §§ 12–14 of paper 9429, the segregated system was adopted. Could the Authors comment on how effective the platform screen doors have been from the points of view of operation and the savings in energy that were forecast.

17. Both 'design and construct' and 'construct only' forms of contract were used in this project. I appreciate that there is some controversy in this area, but I believe we should judge this question according to facts and not according to fashion.

18. In § 79 of Paper 9428 there is an implication that design and construct shortens the overall construction period, and, although this may be true in some cases, I should be interested if the Authors could provide more evidence to prove this point. I notice that the tender preparation for design and construct takes, it is recorded, approximately two months longer than construct only. The time required for the tender evaluations of a design and construct contract has not been given, but I suspect that, overall, an additional four months or more may be required during tendering.

19. In § 81 of Paper 9428, another advantage of design and construct is said to be that it enables contractors to prepare the design to suit their own respective constructional arrangements. It is my experience that this can be equally well incorporated in a construct only contract. For instance, on the Great Belt Eastern Tunnel Project, which is now under way in Denmark, the contractor has had a considerable involvement in design development during the initial stage of his construction contract, as we have prepared the detailed design.

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20. In Paper 9429, concern is expressed about the practicalities of requiring each individual design and construct contractor to prepare his own materials and workmanship specifications, and also about the need for equipment design to precede the civil design, which is another important requirement.

21. All of this, it seems to me, highlights some of the difficulties of 'design and construct', except for this apparent shortening of the programme. In this respect, I think I am right to recall that the phase III programme, which was longer than the extremely fast programme which has been achieved, was based on the budget constraint that Mr Hulme mentioned. In the event, was Mr Hulme able to keep within an average of 5% of the development budget for Singapore. My own view is that, for major infrastructure work in a city, it is difficult to see the advantage of 'design and construct', although 'construct only' does require advance planning which may not always be politically acceptable.

N. Yarwood, *Tony Gee and Partners*

When the construction of the MRT was at its peak, there was a fairly major depression in the original market in civil engineering. Did this happen by luck or by design, or by a little of each?

23. Although there were a number of UK based or UK originated specialist contractors involved with the project, there was a distinct lack of British main contractors on the civils works. Was it because their tenders were too high, or was it that they were less imaginative in the way they worked out the finance?

24. Finally, with regard to the cross-over technique that has been mentioned, I think it is the same technique that has been used for the Hong Kong MTR which has been quite successful, and I assume it has had equal success in Singapore.

K. G. Rippin, *Binnie & Partners, Shanghai*

The Authors have raised some interesting matters with respect to the forms of contract used on the Singapore MRT.

26. The early contracts were prepared on a 'design and construct' basis in order to save time. Contractors' designs were based on general arrangement drawings prepared by the Employer. When tendering, the contractor was required to submit a fixed lump sum price for the total contract, inclusive of detailed architectural works. He also had a responsibility to co-ordinate the work of the electrical and mechanical contractors. The tenderers would presumably have undertaken a considerable amount of preliminary design during the four-month tender period. Even so, it must have been difficult accurately to identify all costs at the time of tender. The Engineer had the power to accept or reject the contractor's detailed designs on which his tender price was based at a later date. Also, the contractor was required to prepare his own materials and workmanship specification, which again the Engineer had the power to accept or reject at a later date. The total amount claimed by contractors in formal claims by the end of the project was $300 million, with an expected payout of $100 million, representing only about 2% of the project cost, which seems a remarkable achievement, and a credit to the contractors' and the Engineer's staff concerned, given the potential that must exist for claims on contracts of this type.

27. The Authors have commented that, on reflection, it would have been preferable for the Engineer to play a more active supervisory role in the co-ordination of civil contracts with electrical and mechanical contracts. When the Engineer enters the arena of co-ordinating contractors' work at a detailed level,

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then the Employer is exposed to risk. Contractors are skilled at co-ordinating work, whereas the Engineer's staff on the whole do not possess the necessary training or experience. One way of eliminating the civil/electrical and mechanical interface might have been to combine the contracts into total civil/electrical/mechanical packages.

28. The Authors note that subcontract nomination was not adopted because the Employer did not want to dilute the main contractor's fundamental responsibilities. However, the description that followed of the system-wide subcontractor who was appointed to carry out work for all main contractors (even those that had not at that stage been engaged) seemed fundamentally to amount to the same thing as subcontract nomination.

29. The MRTC worked up a design and construct document, using the *FIDIC civil conditions* as a base, and adding clauses to cover design and co-ordination responsibilities. Design and construct type contracts, sometimes including civil, mechanical and electrical works as a single package, are becoming more common; yet, to my knowledge, there is not a standard set of internationally accepted conditions of contract currently available for this type of contract.

30. The procedures for managing variation orders is well described by the Authors. The variation originator completed a request sheet, stating the proposal, its purpose and estimated cost, together with a justification questionnaire sheet. This was circulated to various managers for comment, and then went to either the Project Director, the Executive Committee of the MRTC or the MRTC Board for approval, depending on the value of the variation. This procedure appears to suit the carefully considered variation that originates from the design office with plenty of time in hand, but appears to look a bit cumbersome for those variations that arise on site (perhaps originating from a contractor's suggestion for overcoming an immediate difficulty). Could the Authors comment on how well the procedure worked in practice.

M. A. Soubry, *Acer Freeman Fox*

Could the Authors of Paper 9428 say what proportion of costs arose as a result of the civil defence requirements? The entry and egress trial by over 8000 people cited by Mr Copsey in Paper 9429 must have been an awesome piece of organization, and very necessary if such large numbers of people are expected to be accommodated. For what period of time have the station amenities and environmental systems been designed to accommodate such numbers?

32. Purpose-built public shelters are normally designed for smaller numbers of people; such great numbers concentrated at stations might be considered to present an enemy with a potential target, or, at least, to increase the risk that an event will be catastrophic. Do the Authors have any view on this risk? Presumably, the MRT is also provided with suitable modern doors to protect stations where the adjacent tunnels pass under waterways, similar to the London Underground flood doors.

33. For blast-resistant structures to provide the best protection, continuity is very important. Could the Authors explain the structural treatment at the junction between the upper and lower walls where they meet the concourse slab. Fig. 18 of Paper 9429 shows very little continuity.

34. The diaphragm walls outside the in-situ station box are assumed to have contributed to the protection. The two-layer wall will not provide a resistance-to-shock equivalent to a single layer with a thickness equal to the two layers com-

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bined. How was the protective value of the diaphragm wall assessed? Were any explosive tests done on such a combination?

D. R. Gutteridge, *Charles Haswell & Partners*

Paper 9428 mentions that developments at Marina South might eventually pay for two or three MRTs. I have not been to Singapore for about four years. I wonder how much development has taken place at Marina South and if it has affected property values in the central business district. The question of whether or not profits from property development can be used to finance MRTs or underground railways is a recurring one.

36. Paper 9428 also mentions the hump profile with viaduct tracks; I accept the energy benefits from this, but most local people would probably think that aesthetics outweigh these. Mr Copsey has mentioned the importance of alignment and of adopting large radii. I agree with that. However, if one is restricted to small radii, it is far better to locate these near to a station where speeds are low rather than mid-way between stations.

37. Could Mr Shirlaw comment on the settlements that occurred in open-cut construction and tunnel construction, and on whether he thinks they could have been less than actually occurred? Some settlements were substantial.

G. Lance, *Halcrow (currently on secondment to Eurotunnel)*

Were any measurements taken of the movement of the ground either above or adjacent to the tunnels in the Newton area, bearing in mind the location of the canal? Was there any recorded movement of the linings themselves—for example, radially or tangentially—and how did they compare with the theory?

39. What durability monitoring was undertaken of the reinforced concrete structures in place? Were there any permanent measuring points, instruments, probes, etc., placed in the linings to enable half-cell potential measurements to be taken or perhaps to measure the depassivation of the rebar or the rate of any chloride ingress? Alternatively, was the corrosion monitoring done by a visual inspection programme? In one case, the initial lining was ribbed and lagged, and I assume that the in-situ secondary lining was cast directly against the primary lining, without the placing of a debonding member between the two. What was done to control the cracking of the secondary lining, and was any reinforcement put in to allow for control? Also, was there any marked difference in cracking between those sections of secondary lining which contained rebar and those that did not?

40. Mr Copsey has intimated that the best performer with regard to the prevention of water ingress into the pre-cast linings was the hydrophilic-type gasket. Would he comment on any subsequent experiences with the long-term performance of this gasket, and, in particular, on whether or not it maintains a seal after a few years in service?

41. With regard to the civil defence stations, two different types of wall and roof were compared at the time of the design: i.e. a fairly thin wall with an anti-spalling plate cast on the inside, and a thick wall with dense reinforcement including shear links in both directions. Could Mr Copsey comment on the constructibility of these two systems, and which of the two turned out to be the easiest to construct?

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42. What are the wear rates on the small radii curves and have any of these rails been replaced yet?

S. A. Robinson, *Maunsell Consultants Asia*

In § 41 of Paper 9429, the Authors state that the weight of ballasted track used on the viaducts is approximately the same as that of the fixed slab track system used in the tunnels. However, it is significantly higher than that of a track plinth system, which is the likely alternative to ballasted track on the viaduct sections. Therefore, there must have been some additional cost in making the viaduct beams strong enough to carry the extra load. On the other hand, there are many advantages. In addition to those listed (easy maintenance, reduced noise and vibration, ability to tolerate differential settlement), the ballasted track solution has the following advantages.

(*a*) The ballast can accommodate the longitudinal movement that occurs at expansion joints between the viaduct beams and the crossheads.
(*b*) The thickness of the layer of ballast under the sleepers can be allowed to vary quite a lot, so tolerances on viaduct construction are not as critical as may be required with a track plinth design.
(*c*) The extra capacity of the viaduct beams to carry the ballast allowed flexibility during construction, as each beam could accommodate the full weight of another pre-cast beam plus a launching girder.

44. In § 83 of Paper 9429, the Authors refer to the measurement of temperature of thick conrete sections. This was all the more necessary because of the high heat of hydration of the cement available in Singapore. As a result of the monitoring, many pours had to be protected with thermal insulation for several days. Not surprisingly, a great deal of cracking still occurred, particularly in the thick walls and roof slabs of the stations. As the cracks were generally parallel to the main reinforcement, it was decided that it was necessary to pressure grout these cracks, as described in § 88.

45. Grouting was avoided in one section by the use of crack inducers in the outer walls and roof slab, at about 7 m centres. The crack inducer consisted of a notch at the concrete face and a plate cast into the wall at mid-depth. This plate was placed perpendicular to the wall and its width was about 30% of the wall thickness: i.e. a large enough proportion to overcome any variation in concrete tensile strength. Watertightness was achieved with a PVC water bar and a water expansive sealant on each face of the plate. This method worked, in that there were no cracks of any significance except at the crack inducers.

46. A further advantage of the three bearing system (§§ 95 and 96 of Paper 9429) is that it will be more tolerant of variations in the depth of replacement bearings, whenever replacement should prove to be necessary. With the four bearing arrangement, a difference of only 1 mm between the depths of the original and the replacement bearings may lead to significant redistribution of load. As the original bearings had a ± 3 mm tolerance on depth, each replacement bearing would have to be accurately matched with the original before installation, which is a time-consuming process.

47. The design of the depot deck system (Fig. 30 of Paper 9429) uses a precast beam, supported by rubber bearing pads sitting on a small pile cap, typically with two piles. Although this had some advantages, the articulation of the system meant that it was necessary to have a tight tolerance on the location of the heads

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of the driven piles, to avoid significant bending moments in the piles. An arrangement less sensitive in this respect would have eased the problem in driving the 15 000 piles.

48. The Engineer's design contracts required the consultants to produce a complete structural and architectural design within a very short period (§§ 37 and 117 of paper 9429)—about eight months in one case. For the Hong Kong MTR, the consultants were required to produce tender packages in a similar time-scale, but design was far from 100% at that stage. A tender package, for instance, did not include the majority of the reinforcement drawings—these were issued successively during the construction period, to suit the schedule. This system allowed the consultant more time to refine the design, leaving the time-consuming preparation of much of the detail until later. Obviously, many of the tender quantities had to be adjusted, but generally this was not a problem. Do the Authors think that this approach would have been appropriate for the Singapore MRT?

49. The introduction to Paper 9430 has, if anything, understated the speed of construction. At the end of 1986, construction work was being carried out simultaneously at 40 out of the 42 stations, together with their associated sections of tunnel and viaduct.

50. The vital role played by the Singapore Government and its relevant departments (§ 151 of Paper 9430) is correctly emphasized. As with the Hong Kong MTR, the government gave top priority to the MRT project, and minor problems were generally dealt with quickly enough to prevent their becoming major ones. Without this very positive attitude, it would not have been possible to maintain anything like such a fast design and construction schedule.

G. Eastwood, *Mott MacDonald*

The construction of a magnificent mass transit railway system within such a short time, within budget and ahead of programme is a notable achievement for all concerned. One cannot help contrasting the large investment made by the government of a country with one-twentieth of the population of that of the UK, with our own government's policy that public funds should not be made available for such a vital part of our infrastructure, the Channel Tunnel rail link.

52. In §§ 116–119 of Paper 9429, the Authors make a plea for a more reasonable design period and realistic fees for designers. I think the engineers and architects that were involved in the design would echo these sentiments; but the viaduct did not turn out too badly, and I am not sure that things would have been done very differently if more time had been available. Does Mr Potter (Paper 9430) have any comments on or criticsms of the viaduct design?

53. There is one interesting point made in the Papers but not mentioned by Mr Potter, and that is the different viaduct bearing configurations adopted by the designers of different phases of the system. The Authors mentioned the savings in bearing costs that could be achieved by using three rather than four bearings per beam, thus eliminating the high reactions arising from the table leg effect that occurs when the beams try to span diagonally. This is very true. A more significant saving than the saving of bearing costs was in the reduction of crosshead reinforcement required that could be made by using three bearings.

54. However, I should not like to have to replace a bearing placed centrally on a half joint, and we considered that the problems of bearing replacement presented by the three bearing design were fewer than the benefits that would be achieved; for that reason, a four bearing configuration was adopted on phases I and IA.

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D. P. Nicholson, *Ove Arup and Partners*

The very soft marine clay was encountered along many parts of the Singapore Metro alignment. Over Arup and Partners provided a number of contractors with tender and contract design assistance on contracts 104, 105, 105A, 109 and 301, where this clay was encountered. Our design work was mainly on vane strength profiles.

56. Figure 1 shows a profile of vane tests from Newton Station Contract 105A.[1] At depths shallower than 10 m, the vane shear strength is less than 20 kN/m^2. The undrained strength (C_u) design line is also shown in Fig. 1. It plots along the lower bound of the vane strength data. No additional vane correction factors[2] were applied to this line. A typical normally consolidated line assuming $C_u/\sigma'_v = 0{\cdot}25$ is also shown; this shows the soil to be slightly overconsolidated.

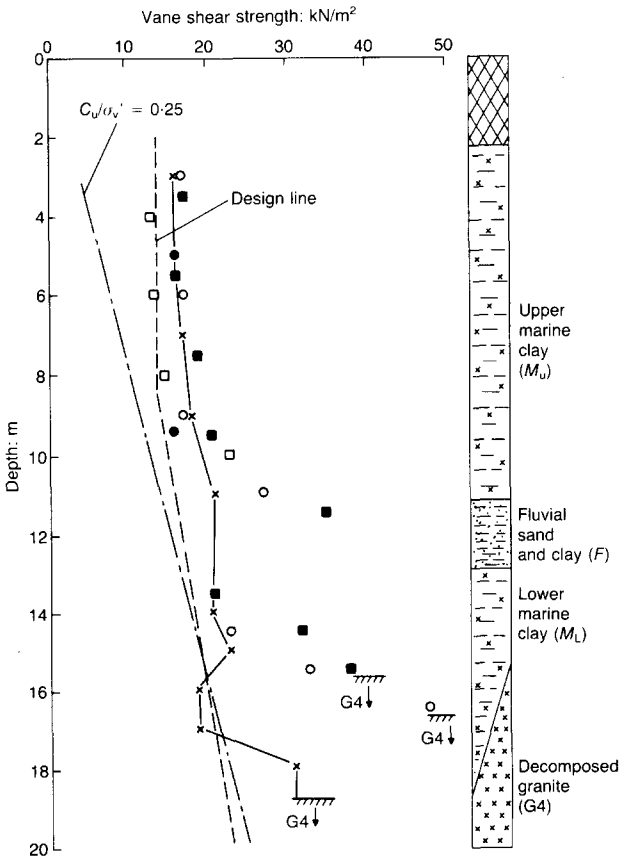


Fig. 1. Newton Station contract 105A: typical vane strength profile (vane dimensions 130 × 65 mm)

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57. In Paper 9430, the marine clay vane strength profile for Bugis Station is reported in Fig. 3. This shows vane strengths greater than 25 kN/m^2 from 0 to 10 m. The mean vane strength line is also shown. Could the Authors answer the following questions?

(*a*) Why were the vane strengths at Bugis Station higher than those at Newton Station and other MRT Stations?
(*b*) Can this difference in strength be explained by the surcharge applied from buildings in the area?
(*c*) Do the oedometer preconsolidation pressures reflect higher over-consolidation ratios?
(*d*) Was a vane correction factor[2] applied to the mean vane strength line to obtain a design line?
(*e*) Were any anisotropically consolidated triaxial undrained compression or extension tests carried out for any of the MRTC contracts to enable a SHANSEP approach to be adopted? What results were obtained?

58. Figures 5–7 of Paper 9430 refer to temporary retaining walls where failures were encountered. It would be very useful to have additional details on the background to these failures, so that the same problems can be avoided in future.

59. In the case of Orchard Station, Ove Arup and Partners were called in to assist with remedial work. The original nail slope design was based on total stress parameters for the decomposed granite (G4). Our revised design was based on effective stress parameters. Drainage was also installed to lower the groundwater table in the nailed zone. In the case of City Hall, it appeared that softening occurred in the S3 behind the gound-anchor spreader beam.

60. Figure 22 of Paper 9430 gives details of the ground surface settlements outside Newton Station. About 100 mm of the movement was associated with consolidation of the marine clay, resulting from a lowering of the groundwater in the underlying decomposed granite. Details of the piezometer responses are reported in reference 1. At Newton Station it was necessary to pump from inside the station box to avoid hydraulic uplift problems with the overlying marine clay during excavation.

61. Could the Authors explain what dewatering was undertaken at Bugis Station, and what piezometer response was observed in the Old Alluvium?

62. Figure 9 of Paper 9430 shows the use of chemico-lime piles. Could the Authors provide additional details of this system and of what improvement in soil strength was achieved?

63. A maximum overall leakage rate into the tunnels of 5 ml m^{-2} h^{-1} was specified to reduce consolidation attributable to drainage into the tunnels. Fig. 23 of Paper 9430 identifies the surface consolidation settlements which occurred up to completion of the permanent lining. Did the subsequent surface monitoring records show that no further consolidation settlement occurred?

64. The main foundation design difficulties with design and construct contracts at tender time are: lack of reliable site investigation data; lack of time to develop cost-effective schemes. In my opinion, it is very important for the contractor to state the geological sequence (contoured plans are best, not just a long section) and the soil design parameters adopted in the tender. The Client's engineers can then agree to these before signing the contract. This information then forms the basis for the Clause 12 unforeseen conditions. Do the Authors have any views on this problem, based on their MRTC claims experiences?

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N. Yarwood, *Tony Gee and Partners*

We were involved in the development of the construction equipment for eight of the 13 contracts. Therefore, I should like to ask one or two questions about the design in relation to its constructibility.

66. In Paper 9430, it is stated that the predictions of the ground movement were reasonably well met by the actual ground movements. How accurately were they met and, also, if people were involved, what particular programmes were actually used to establish these ground movements?

67. With regard to the superstructure construction, an enormous variety of methods were used in the construction. Other contractors possibly spent a great deal more money initially in buying equipment which then paid enormous dividends later on.

68. An enormous difference between the two sets of contracts existed: the 300 series contracts and the remainder. The 300 contracts had beams that ran up to 260 t. In one case on contract 305, we were faced with a beam that weighed 270 t, which actually had an end reaction of 175 t because it was asymmetric, giving us the problem of placing a beam the equivalent of 350 t in weight. With hindsight, do the Authors know whether it was the right decision to go for beams that were very much heavier in the 300 series contracts as opposed to the earlier contracts which used a greater amount of in-situ design. Our policy for lifting and towering these massive loads was to use hydraulics wherever possible: firstly, they are far less bulky; secondly, they are cheaper and more reliable.

69. We had over 3000 beams to place in the whole series, and these probably involved 15 000 lifting operations of over 155 t. It was inevitable that sooner or later one or two things would go wrong, especially as beams had to be delivered down slopes. There was one incident in which an electric drive, which does not have the same ability as the hydraulics to provide braking, failed, with fairly disastrous consequences.

70. There were problems with the very large gantries which were used to place the very heavy beams, as these were very difficult to manipulate over the fairly short spans. However, I have to say that the method of using launching gantries for pre-cast beams was extremely successful, and some of the construction rates that were achived by these gantries were quite extraordinary. In one case I know of, four beams were placed in one day, including the launch.

71. In may view, the Authors of the Papers, the people who had the vision to conceive this enormous system, and those who built it in such a short time, on what is really quite a moderate budget by modern mega-project standards, deserve to be congratulated.

A. J. Howard, *Mott MacDonald*

If there is no research and development provision in such a large project, who else is going to pay for research and development? I am sure that the MRTC contributed something. Perhaps the Authors might comment on that.

73. The MRTC was faced with quite a large number of contractors' design consultants. The Papers do not mention who they were. It would be interesting to know how many contractors did the design themselves.

74. For the design and construct contracts, the contractors had to mobilize not only their own staff but the Consultants' staff as well. How well did that go? Was time lost in actually getting a new team of people mobilized and working? How were the relationships with all of these design contractors? I was surprised to

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see the comment in § 116 of Paper 9429 that, in practice, contractors 'selected by international tender were intent on providing minimum standards subject only to compliance with the MRT Corporation's performance specifications'. It is a view that I perhaps had and felt cynical about having. To what extent was this a real problem?

75. I should like to comment on the effect of changes in mechanical and electrical design which have occurred all the way through the project. When the phase 3 study was carried out, all of the electrical and mechanical elements were first, and at the bottom was civil and structural design. I think all civil engineers who have designed any complex works have had this problem. However, getting the mechanical and electrical elements right and fixed is one of the most difficult problems. Should the design have started the other way round?

76. I was concerned to read about the acid in the ground. I know of another situation where sulphuric acid generated in the ground is eating cast-iron tunnel linings. Trains passing through the tunnels could pump air into the ground, thereby enabling the reaction to continue. The movement of groundwater may also bring additional oxygen. I see that protection has been provided but this must be a matter of ongoing concern.

77. In § 66 of Paper 9430, it is stated that the settlement in good ground was sometimes greater than the settlement in bad ground; this seems to be very unusual. Did the Authors really mean that, because the ground was better, the contractors were less punctilious in their observance of good tunnelling practice which led to greater settlement; whereas in the bad ground, everything was tightened up? In Fig. 23 of Paper 9430, the settlement has reached 100 mm and is still continuing rather rapidly. Has settlement since stopped?

D. Stewart, *Stewart International Co. Ltd, Taipei*

With reference to Paper 9430, some mention is made of difficulties experienced during the tunnelling work (see §§ 44 and 64). When tunnelling contracts experience difficulties, delays are often inevitable unless further resources are employed. Assuming that the contractor regards the event as Clause 12 material, which is usually the case, the poject encounters something of a dilemma.

79. Under the FIDIC terms of contract, the contractor cannot resource the job on the basis of the extensions of time he may hope to receive at some time in the future, when the Engineer has evaluated the claim and given his decision. The Engineer may not be able to give a decision until the tunnelling has been completed. An example which demonstrates this point was given by the late Charles Haswell,[3] when he observed in a presentation to the British Tunnelling Society that London clay was known to contain pockets of sand. If one pocket was encountered on a half-mile drive, that was the contractor's risk; if three pockets were encountered, he suggested that that may well constitute a Clause 12 claim. The dilemma I wish to highlight applies equally to the contractor and the owner. The contractor has to decide on a course of action without any certainty that the claim will be successful, and the owner is left to wait until the work is completed to hear the Engineer's ruling on his liability for increased costs. It is both the scale of the increased costs and the almost certainty that, in tunnelling, increased resources are the only solution that have led to the demise of so many of the well-known British tunnelling companies in the past two decades.

80. Possible solutions have been advanced world-wide to improve contract practices in tunnelling, including reference conditions, on the job mediation and

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peer review panels. Do the Authors have any recommendations on how to improve contract practice for tunnelling works.

G. Crook, *Godfrey Crook Associates*

Is the Mass Transit system going to be linked to Changi Airport? I find it extraordinary that such major transport systems are not linked to the international airport—the principal way of arriving in a country. In Hong Kong, also, it was decided for reasons which are unclear that Kaitak airport is not to be linked to the Mass Transit railway.

Mr Hulme

Planning in Singapore, as *Mr Howard* points out, was straightforward in that the government, because it was in the process of providing housing for the population on a very large scale, was able to direct the residential locations of the mass perhaps to a greater extent than would be possible in an 'older' country. During the 1979 study, the question was asked of why the MRT was not going as far north as Yishun new town. When it was pointed out that it was still a green field site, the reply was: 'How many people do you need to be at Yishun to make it worthwhile to extend the MRT—50 000, 75 000?'. On confirming that 75 000 would do very well, the Housing Development Board promptly agreed and put a new town there, with a population of 75 000. It was a significant advantage that most of the industrial parks and housing estates were still being built and could be planned as the classic string of pearls joined up by the railway.

83. There is potential traffic in the Bukit Timah Road corridor, but people living there tend to be fairly affluent and would perhaps strongly resist being asked to leave their cars at home; whereas in the new towns, the expectation was that many of the inhabitants would not have any alternative means of transport. Therefore, the emphasis was put on joining these up first, and the Bukit Timah corridor has been left to be served by cars and buses. The choice was to serve the less well off sections of Singapore and to leave the more affluent sections to sort out their own transportation problems. This answers to some extent *Mr Malcolm's* point.

84. *Mr Howard* and *Mr Yarwood* both ask what effect the economic downturn had on planning. It may not have been good planning entirely, because the decision to proceed was made in 1982 and it is doubtful if anybody at that time appreciated how severe the depression would be. However, by the time of the start of award of contracts in late 1983, the recession was severe and the early tenders reflected the keenness of contractors to 'get a part of the action'. It is difficult to say what the long-term effect on contractors is. Clearly, if the workforce is underused and the contractor wants to keep it together, then it may be worthwhile, under those circumstances, to take work at very low profit margins. The question of whether or not it every pays to even risk losing money just to keep the workforce together is debatable. From the Coporation's point of view, when it became evident in late 1983 and early 1984 that there was a very serious recession, the decision was made to accelerate the letting of the rest of the contracts. In a way, the tail then wagged the dog. It was not entirely the desire to reduce the overall programme from nine to seven years that forced the decision; it was the keenness to let the contracts while contractors were short of work and would therefore submit very competitive tenders, and, of course, it would ensure world-wide inter-

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est. If there had been other competing projects—if Hong Kong had been going on longer or Taipei had started earlier—Singapore would probably not have received the interest it did. However, there was a very good response to the invitation to submit tenders, and, owing to both the world-wide competition and the world-wide recession, this did result in some very low prices for phase I. Prices tended to creep up in comparison with the Engineer's estimate as the project went ahead into phase II, but overall the tenders came in lower than had been anticipated.

85. Contractors must say whether or not that was a good move or not. A great deal of risk was, quite intentionally, put on to contractors, and the contract documents made that clear. The *FIDIC Conditions of contract* was the basis for the documents, but the Corporation modified them by altering words in clauses. While a contractor should always read every clause very carefully when he is tendering, it is asking a great deal to do it in the time available. It may have been better to have taken *FIDIC* and deleted whole clauses, and then to have inserted new ones, so that contractors would know where the changes had been made and could therefore make due allowance for them. But the risk was there, contractors took it willingly and there was a Clause 12 to allow for unforeseen conditions. The Corporation's stated position has always been to pay whatever amounts the contractors are fairly entitled to. However, between what contractors feel they are fairly entitled to and what the clients believe is a fair payment usually shows a very large gap. Final payments are very approximately a third of the original claims submitted by contractors.

86. With regard to the risks placed on contractors in future government legislation, the only really serious one was that dealing with compressed air working. There was no legislation in Singapore, and the tender documents required that the contractor should allow for working to conditions similar to those in Hong Kong and should also be prepared to comply with the *CIRIA Medical code of practice for work in compressed air*. Contractors were also advised that when the Singapore legislation was enacted, they would have to comply with it and it would not be accepted as the basis of a claim. It seems very harsh, but in actual fact the legislation did follow very closely the Hong Kong model, and overall this method of specification was not found to be particularly onerous.

87. As *Mr Malcolm* points out, there was a very good legacy from the earlier colonial days, but it has to be recognized that the early days in the birth of Singapore were quite difficult. Singapore was initially part of Malaysia, but after the two agreed it would be better for them to separate, it became an independent state in 1965. Singapore then experienced a tremendous population burst: the pre-war population was between half and three-quarters of a million, whereas in 1965 it was approaching two million and expanding rapidly. The Government of Singapore was faced with a very difficult situation. The UK Government gave rather short notice that it was going to withdraw all military personnel and establishments, and this meant that the government had to develop a whole new industrial basis for the economy. Looking back at what has been achieved in 25 years, it is quite incredible that the whole economy was restructured without getting into debt—quite the opposite, in fact, as the overseas reserves of Singapore in relation to the population must be some of the largest in the world. Singapore has well and truly improved on what was a good basis.

88. *Mr Elliott* raises the question of the relative merits of design and construct against engineer design. Design and construct is a little quicker; however, in a fairly detailed investigation made before the start of phase II, the conclusion was

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reached that the advantage was a matter of months not of years, and it was considered that it did not overcome some of the other problems. It is more difficult to ensure a uniform product with design and construct because there are different contractors with a variety of ideas working on the various sections. Sometimes this has a plus side. The stations in the underground sections were intended to be individual, so that people would be able to recognize their destination, and to be varied and interesting; a design and construct contract meant that virtually every station had a different architect. Above ground this was not thought so necessary and, in fact, the consultants advised and the Corporation accepted that it was better to have a uniform theme over a number of stations. Passengers can orientate themselves by the surrounding topography and they do not need the stations to be quite so individual. So the advantage of design and construct was experienced underground, but there were other problems, such as the difficulty of ascertaining a fair basis of payment. In the end, a general guideline was drawn up that if the contractor, having made a responsible effort at design, could not fit all the required facilities into the box shown on the tender drawings then that was a fair basis on which to say that additional payment was due. However, it is not easy to separate what is reasonable design development—which the contractor can be expected to have allowed for—from what is a change or a variation—which has to be paid for. It was complicated because the NFPA 130 code of North America, dealing basically with fire and life safety design, was introduced as a criterion at a late stage. It was accepted that this was a variation in the Corporation's requirements, but it was not easy to find a way of separating the effects of that from what a reasonable contractor should have provided in the way of escape services.

89. Although there are advantages in design and construct, and the contractor can direct the design into a form that suits his own particular expertise, it has quite a few other drawbacks. From experience in Singapore, the Engineer design allows the final product to be more nearly that which the client wants.

90. The other lesson learnt is that of having a clause or provision for value engineering. It is beneficial if the contractor is interested and wants to put in his ideas. The only way to achieve that co-operation is to give the contractor a financial incentive: if he can come up with a good idea and save money then he ought to participate in that saving. That, at least, is one means by which interaction between the contractor and the designer can be fostered.

91. As mentioned by *Mr Elliott*, the use of platform screen doors was looked into during the *Phase III preliminary engineering design design study* in 1980. At that time, Lille was introducing screen doors, and Leningrad was the only other system with screen doors but for a rather different purpose: they were meant to keep the cold out rather than to keep it in. One member of the team did make a trip to Leningrad to have a look at the operation there. Of course, one of the problems is the need to have very accurate train stopping if the platform screen doors are to function successfully. This is achieved in Leningrad by a system whereby the train enters the station very slowly, but this is hardly acceptable on a modern system. However, signalling has shown tremendous improvements, even since the early days of the Hong Kong MTR, so that it became clear that stopping within about plus or minus half-a-metre was quite practical in present day automatic train operation. That overcame one problem with the screen doors; the other problem was the doors' reliability, which has been one of the pleasant surprises in Singapore. Phase I opened in November–December 1987, and most of the underground stations were in that phase. Therefore, the doors have now been

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operating for two years and there have been no major problems. The trolleys—the doors are hung from a top support system—have been modified in that period, but there have been no major breakdowns of the equipment. Train stopping was a problem initially, but with the present revised software, the stopping accuracy is now approaching the specified 99·79% stops within plus or minus half-a-metre.

92. As far as can be determined, the savings which were expected in air-conditioning costs in stations have been achieved: i.e. up to 50% of the costs that would have been incurred if the open system had been installed, when there would have been a tendency for the cold air in stations to be drawn out by the trains as they left, and for the hot tunnel air to be introduced into the station as the trains entered. The doors have also been reassuring, in a country where there is no tradition of railway travel, in that they give some form of security in the station underground, although the crowds are not sufficiently great at present to pose any safety hazard. The long-term maintenance costs and maintenance problems are still unknown, but up to now nothing serious has become apparent.

93. The traction power supply is energy efficient because of the use of regenerative braking on the trains. The hump profile has been incorporated on the underground section only, although originally it was also applied on the above-ground sections. On review it was considered that, as regenerative braking had already been incorporated, the expenditure involved in raising the viaduct stations above the general level of the track was not worthwhile. One of the operating lessons learnt is that it would have been worthwhile to retain the hump throughout the system because, although energy savings are not so significant, there are braking and acceleration benefits and, in particular, a braking effect without having to use brakes. In heavy rain in Singapore, it has now been found necessary to reduce the braking rate of trains to ensure stopping accuracy, so that travel times are slower when the tracks are wet. With the acceleration out of the station and down the hump there is also an improvement in travel times. In any extensions, the use of a hump profile above ground as well as below ground would be re-examined. The belief had been that, with regenerative braking, humps would not be needed. In fact, they are needed, but for reasons other than energy conservation.

94. Were project costs within 5% of the annual development budget? A brief review indicates that costs generally were contained within the 5% of the annual development budget, but this was assisted by the fact that the increase in the gross national product had been greater than allowed for. Annual expenditure was, therefore, greater than originally estimated because the project had been accelerated. However, because the growth of Singapore has been greater than allowed for in 1979–80, costs have kept just about to that 5% target.

95. The variation procedure, as *Mr Rippin* points out, requires time, although much depends on whether or not the originator sees his proposal through. For urgent site variations, both the senior resident engineer and the construction manager had financial limits within which instructions could be issued to the contractor immediately, and these were then automatically authorized retrospectively every month by the 'request for variation' mechanism.

96. *Mr Malcolm* questions the lack of planning for an extension to Malaysia. The alignment round the north of the island has very carefully kept open the option of a spur to go over the causeway. Provision of a major interchange on the Singapore side of the causeway with the Malayan railway and with the road system has also been examined. The proposal has been to have a road, main line and MRT interchange at the causeway, so that passengers coming from Malaysia

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could transfer very conveniently, and it seems likely that something similar will come to fruition. There are many problems to be overcome here, not least is that of working between two nations, as it is not easy to try and reconcile the way in which two sovereign nations perceive the advantages for their own economies. Of course, very serious thought is being given to a second connection, but the question of whether it should be a widened causeway or whether an independent one is again giving rise to a considerable debate. Johore probably favours a second crossing separate from the causeway because that would open up for development other areas of Johore. The Malaysian state may see it a different way, wanting the connection to benefit the whole of Malaysia and not Johore state alone. For Singapore to extend its MRT acoss the causeway would therefore be quite difficult. There would be obvious customs problems: at the moment, there are Malaysian customs on the Malaysian side of the causeway and Singapore customs are on the Singapore side. There has been talk of trying to combine these but questions of sovereignty do come into it. Clearly, there is every benefit to be gained from providing at least a very good and efficient interchange at the causeway so that travellers coming down the peninsula can get to any part of Singapore very quickly and conveniently, and, it is to be hoped, by the rapid transit system.

97. In reply to *Mr Soubry's* query on civil defence, I would emphasize that a full-scale test was conducted recently by the civil defence forces of Singapore, in which they mobilized all the surrounding inhabitants. About 8000 people were in the station to test everything out before it was opened to traffic. Civil defence has been taken very seriously and realistic practices have been conducted.

98. *Mr Howard*, in § 72, expresses concern that there had been no research and development. It is correct that research and development was not part of the brief but, inevitably, there has been some development; it cannot be avoided. The viaduct concept with the pre-cast beams and the gantry launching was tried out in Melbourne, and Singapore was a further improvement on the original concept.

99. Similarly, on the track, the floating slab track performed very well at damping out vibrations, but it caused an increase in noise levels in the tunnels, and this has been looked into. After a significant amount of work, a more resilient type of pad, which can be put under the rail fastening, has been developed. This has resulted in a significant reduction in noise levels: i.e. up to 5 and 6 dBA.

100. *Mr Lance* raises the question of rail wear. Rail wear is minimal, even now after the system has been running for two years: this is partly good track design on the civil side and partly good design of the bogies on the rolling stock. They are not self-steering bogies but they have some degree of self-steering which has obviously kept them well centred on the track, and the two together have performed very satisfactorily.

101. *Mr Crook* expresses surprise that the mass transit has not been extended out to Changi Airport. The viability of such an extension has been studied three times, the route out to the airport is safeguarded and there are stub ends on the viaduct near Tanah Merah Station ready for the connection if it is ever implemented. However, none of the studies has ever shown an economic case sufficiently strong. Changi is an international airport without the short-haul traffic that is found in European or American airports. The Kuala Lumpur shuttle is probably the only real equivalent. The airport is tremendously busy from about 5 pm to 10 pm every evening, but during the rest of the day the traffic is not particularly heavy. The bulk of the traffic the mass transit would take would therefore be in the evening. It would also comprise long-haul travellers with a great deal of luggage,

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who want mostly just to get into a taxi and be taken to their destination without any problems. The other consideration is that the taxis in Singapore are phenomenal. There are so many taxis lined up waiting for traffic at Changi Airport that a wait of more than five minutes for one, even at the peak hour, is very unusual. Competition from other transportation modes is therefore strong, and it is likely to be sometime before the MRT goes out to the airport.

102. *Mr Robinson*, in § 48, asks about the merit of inviting tenders before the design is complete to allow more time for consultants to prepare details. As noted in Paper 9429, the acceleration of Phase II construction by two years resulted in a tight design schedule, and it tended to force the consultants to adopt over-conservative concepts. However, the Authors believe that the decision to produce a complete design before proceeding to invite tenders had the following advantages.

(*a*) With a definite contract period, the consultants were able to assess their manpower needs and overheads better; this helped to lower the design contract prices.

(*b*) The contractors were in a better position to price the bills of quantities, as the details were complete and the quantities were more or less final. This is reflected in low variations/claims received in the phase IIA/IIB contracts.

(*c*) No delays resulting from the design process were experienced in the construction. In practice, on a number of occasions in Singapore, programme constraints meant that tenders had to be called even though the design was not 100% complete.

103. Wherever the need for refinement was identified after the final design, the consultants were asked to carry out the refinement as extra work (e.g. Bugis, Lavender and Marina Bay stations). However, there were disputes at times as to whether the refinement of an over-conservative design should be considered as extra work or to the designer's account. In the last phase (Phase IIC), the design contract allowed for the consultant's key design personnel to be seconded to the site during the construction period. This provision enabled the consultants to refine/modify the design where necessary, without having to argue out whether or not it was an extra task. This arrangement resulted in very satisfactory results in terms of both quality of the design and design management.

104. In conclusion, the Authors feel that the practice of completing the design before proceeding to invite tenders has many avantages. However, as noted in § 117 of Paper 9429, it is essential to plan forward effectively to allow a reasonable minimum design period.

105. *Mr Stewart's* question opens up a vast field to which only a brief reply is possible. Improved contract documents would help, and the CIRIA Report 79 points the way. Too many clients still try to impose their own pet conditions on contractors without having any comprehension as to their real effect. Adequate time to prepare documents by genuinely experienced engineers would remove some elementary ambiguities which usually end up as claims. Reference conditions, or something similar, would allow the consultant who has had the time and resources to study the underground conditions to indicate his conclusions to tenderers and could be an improvement—if clients were willing to pay for this service. Deletion of unforeseen condition clauses, beloved solution of clients, must result in either higher costs or breach of contract claims. Contractors should be

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prepared to submit well substantiated claims immediately on occurrence, and not prevaricate in the hope of later blaming all their deficiencies on the Engineer/Client. The list is almost endless.

106. *Mr Nicholson*, in § 64, suggests standards for the presentation by the contractor of his geological assumptions at the time of tender, and this seems to me to be a most worthwhile proposal in reducing subjective argument during the contract.

107. One particular action based on the Singapore experience is for tender evaluation to be more positive: e.g. tenders for four critical drives, using only one shield, should not be accepted by the client, nor should unsuitable lining proposals. The client must be ready to reject these and other dubious submissions even if it means the acceptance of a higher tender price initially.

108. A general question is that concerning the effect which the MRT has had on car traffic in Singapore. That is very difficult to elucidate because it is masked by the various traffic control systems already in place. Singapore is about the only city in the world with an area licensing scheme which is fully operational: to enter the central business district between 7.30 and 10.15 a.m. requires the purchase of a permit which costs $3 for a private car and $6 for a company car. Therefore, there is pricing of the traffic space. A similar scheme for the evening has been introduced recently, although quite unusually it is still people entering the central business district who pay. The logic is to keep people out of the way so that those in the central business district can get home. However, this type of scheme does mask the effect of the introduction of another transit system. All that the latter has achieved so far is to just about stabilize the growth of car ownership, which had been increasing despite being a very expensive luxury. The price of a car in England is approximately doubled in Singapore, and an annual licence fee of some S$4000 (about £1500) is not unusual. Last year, with the various traffic control measures and perhaps the introduction of the MRT, car ownership levelled off, but this situation may not continue. It is very difficult to win people away from their car. Ticketing between buses and the railway was introduced experimentally in February 1991. This again is an inducement to people to use public transport all the way, and this may be a significant factor because most people still have to get a bus to the station, use the train, and then get a bus to their destination. A through ticket which shows some saving financially will perhaps persuade people to use the MRT system, but only time will tell.

109. *Mr Nicholson's* constructive suggestion in § 64 that the contractor should state the geological sequence and soil design parameters adopted in his tender is welcome. The better and more detailed the information is that is submitted by contractors at the tender stage, the more straightforward the evaluation of claims becomes later. Before award, however, clear instruction in the tender documents of what is required of the contractor and careful examination by the Engineer of the resulting submission are necessary.

Mr Copsey

With regard to *Mr Soubry's* comments on the civil defence asspects of the project, it is fair to say that generally the passsengers are not aware of the civil defence construction or of the potential use of the stations for civil defence purposes. Apart from the civil defence logo at the station entrance, there is little to distinguish a station designed for civil defence. There is also little impact on the normal operations of the railway. One particularly interesting aspect relates to the blast doors

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that go across the tracks at each end of the CD station: a key is inserted in each door to lock it in its open position; the lock mechanism is connected to the signalling system, and if the key is not in place the signalling system will prevent train movement.

111. A station is converted to civil defence mode by the closing and sealing of the blast doors across the tracks at station ends and at the station entrances. For the latter, architectural panels in the walls, ceiling and floor are removed to expose the doors and their tracks. Several trials have been held and we have found that it typically takes a maximum of four hours to convert the station to CD mode.

112. In civil defence conditions it is intended to stop a train at each platform to given increased seating capacity. Special ventilation has been installed so that feeders can be brought down from behind the platform ceiling and fed into the train. The improved ventilation in the trains makes them suitable for hospital use.

113. *Mr Soubry* asks what degree of threat would initiate a transfer to CD mode. Various scenarios have been examined, but, as the questioner will appreciate, such matters are outside the purview of the Mass Rapid Transit Corporation. The screen doors have not been designed for gas tightness. The blast doors across the tracks and at station entrances, when fully closed, enable an overpressure to be maintained inside the station. Furthermore, as previously mentioned, there is likely to be a train in each platform which requires the screen doors to be left open. The tunnels could be used in CD mode but would be a last resort. The blast doors across the tracks isolate the tunnels but a bypass door has been provided so that access is possible. The tunnels would provide a second but lower protection to the station areas.

114. The railway facilities have been tried and tested, and to some extent this applies to the civil defence facilities. All the CD findings were designed for shock loads, and many have been tested to simulated shock loads. The operation of civil defence installations has been tested where practicable.

115. The query *Mr Gutteridge* raises on the suitability of hump profiles on the elevated station is one involving aesthetics. The raising of the viaduct level at stations and the lifting of stations 2–3 m above the general track level has not only a cost penalty but is also considered to have a significant aesthetic penalty. It is acknowledged that aesthetics is subjective and that a case can be made for hump profiles along viaduct sections.

116. A minimum radii of 275 m was achieved on the running line. This radius occurred at three locations, and two of these were close to the ends of underground stations. A train accelerating from a station, especially one with a hump profile, can reach a speed of approximately 60 km/h before the end of the train passes the length of the platform. Any curve immediately after or before a station will be subject to trains travelling at reasonable speeds, and it is worthwhile to attempt to achieve the desirable minimum radius throughout the alignment, irrespective of position relative to stations.

117. In response to *Mr Lance's* question on the stability of tunnel linings, the tunnnels at Newton Circus were influenced by substantial settlement from the construction of Newton Station, and it was impossible to distinguish stability movements from movements arising from the station works. Elsewhere along the alignment, measurements of deflexion of the tunnels were taken, and those in marine clay behaved considerably better than expected. A distributional coefficient of 0·7 was assumed in the tunnel design. On contract 104, the circle joint bolts were stressed in order to give an improved interlock between one ring and

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the next, and to increase the flexural stiffness of the lining. Such measures were deemed necessary to cater for the distortion of these linings in marine clay.

118. On the measurements that we carried out, we found that the distortional coefficient was 0·8–0·85. Very little deflexion on the linings in the marine clay took place, typically 5–10 mm. On the other hand, linings in completely decomposed granite (G4) deflected up to 20 mm fom ground loadings.

119. Long-term monitoring of the linings for attack from aggressive grounds consists at present of water sampling, both of water seeping into the tunnels through the joints between segments and of water from outside the tunnel. The linings are well grouted and therefore the pH of groundwater is generally high. The concern is that, in places, the bacteria in the ground, which were principally aerobic, will manufacture acids before the full water pressure is restored and has replaced all air in the ground. It is intended that a series of cores will be drilled through the body of the tunnel segments to check on loss of passivation and to show any ingress of sulphate or chloride ions. No half-cell measurements have been carried out or are proposed.

120. Nominal reinforcement was placed in all in-situ linings and very little cracking has occurred. Virtually all the linings contain silica fume, so there was little opportunity to compare linings with and without fume.

121. In response to *Mr Lance's* question on the long-term performance of the waterproofing system used on contracts 107B and 301, the system has now been in place for three years and is still proving very satisfactory. The waterproofing material, which consists of a waterproof but water-expansive resin, has been on the market in Japan for 10 years and has some track record. The MRT Corporation is very satisfied with the waterproofing system, although it is understood that there have been problems elsewhere with a similar material, which were caused by damage during segment erection. On segments used in contracts 107B and 301, the waterproofing strip was recessed into the segment so that it did not protrude, virtually eliminating risk of damage during erection. The expansive nature of the material was relied on to create the pressure to seal the tunnel joints.

122. There were alternatives for the design of the walls and the roof of civil defence structures: one was to have a certain thickness of concrete with a steel spalling plate on the interior; the second was to have an additional 400 mm of concrete without the spalling plate. Selection was mixed, some designers opting for a spalling plate and others for the extra concrete. It could not be concluded which was the more economical method. Certainly the spalling plate arrangement is more complicated, with hooked reinforcement bars welded to the plate and debonded over part of their length.

123. *Mr Howard* asks which contractors designed the structures themselves. From the 11 design and construct contracts on phase I of the project, two contractors used their own in-house expertise in the design, namely Kajima (contract 107) and Ohbayashi/Okumura (contract 109), and the other contractors employed consultants for their design.

124. The mobilization time for design teams was variable. For some contracts there was significant delay before an acceptable design team could be put together. The quality of the design in design and construct contracts was sometimes unacceptable, and it was difficult on occasion to demonstrate to designers that designs needed to be revised. The MRT Corporation was dealing with international firms, Japanese designers, all of whom had different definitions of good practice; there were bound to be differences of design method. Relationships were good, but a lot

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of strong engineering discussion was necessary before many a design issue was resolved.

125. With regard to *Mr Howard's* question on segment durability, potential problems from groundwater can be minimized if water ingress is not allowed to run down the tunnel lining, spreading out as it does so, and allowing any salts in the water to permeate into the face of the tunnel lining. In Singapore, internal pointing and caulking have been carried out where necessary to control leakage, such that any water entering the tunnel flows immediately into the drainage channels beside the track and is not given the opportunity to spread across the internal face of the segments.

Mr Potter

The cross-section profile of the beams was fairly complex, but aesthetically the beams were very pleasing and achieved what the designer originally envisaged in his conceptual design. The viaducts have blended well into the environment and, in general, no disapproving comments about them have been received. The main engineering criticism has been the half joints and the problem of replacing the bearings, as pointed out by *Mr Eastwood.*

127. A successful test has been carried out on site for removing the single centre bearing of a three-bearing system. The lesson learnt from the test was that the beam has to be securely wedged to prevent any longitudinal or lateral movement once it is lifted off the original bearing. Any slight movement will result in misalignment of the bearing plates' bolt holes, which will prevent fitting the replacement bolts when inserting the replacement bearing.

128. With respect to the standard of workmanship for the viaduct construction, the finished product has overall been very uniform. The majority of viaduct contractors were Japanese, with some in joint venture with Singapore contractors. However, Korean, Taiwan, French and Hong Kong contractors also manufactured the beams for the viaduct contracts. The uniform standard achieved has been mainly due to the use of steel forms for all beam and column construction. Cycle times for beam construction were remarkebly consistent, with most contractors achieving 11–12 day cycle times, except for the two French contractors who used more sophisticated precasting methods to achieve cycle times of seven days.

129. Another point to make on the workmanship was that the MRT Corporation provided full supervision in the casting yards which made a substantial contribution to the attainment of quality. The very good workability of the concrete during placement in the beams was also a major factor; this was achieved because the batching plants were located in the casting yards, which resulted in very good quality control.

130. I agree with the point which *Mr Yarwood* raises in respect of the heavier beams used on the phase IIA viaduct, and, with hindsight, the phase IIA designers could have avoided their use in most cases. In fact, one phase IIA contractor redesigned an in-situ bridge, which had a centre span with two drop-in heavyweight beams, so that he could use his standard beams and therefore avoid providing a strengthened gantry just for the placement of two heavy beams.

131. It is pointed out in Paper 9430 that the consultant designers for phases IA and IIB overcame the problem of providing non-standard beams by enlarging the crossheads to avoid any increase in precast beam spans. However, the Phase IIA consultant generally designed uniform crossheads which resulted in longer, heavier beams in places.

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132. The lessons learnt on the project indicate that the use of non-standard beams for just a few spans should be avoided, as the cost increase for travelling gantries and beam forms are not justified.

Mr Shirlaw

Mr Malcolm asks about dewatering around the tunnels as a method of avoiding problems resulting from aggressive groundwater. Many of the soils in Singapore are highly compressible. Water contents of 200–500% are typical in the organic estuarine clays; while the soft marine clay typically has a water content of between 65% and 80%. So any attempt to reduce the problem of the agressiveness of the groundwater by deliberate dewatering would lead to an even more severe problem of ground settlement. On one contract, dewatering during construction led to a 9 m loss of head outside the excavation. At the same time, there was nearly 600 mm of consolidation settlement. This suggests that deliberate dewatering down to tunnel invert level could have resulted in settlements of 1–2 m.

134. In § 37, *Mr Gutteridge* asks about settlements. After building the system, more is known about controlling settlements than before the start. Unfortunately, it is not possible here to give a comprehensive answer to the question. The range of ground conditions encountered in Singapore was wide, including: soft clays; stiff clays; well-graded sands; poorly graded sands; granites and granodiorites (in six weathering grades); sedimentary rocks (in six weathering grades); colluvial soils; stiff boulder clay; fill; an old coral reef; mixed faces of the different materials listed here.

135. One factor in controlling settlements is to choose the most appropriate tunnelling method for particular type of ground. Ground stabilization can be considered as an ancillary to the tunnelling method.

136. Tunnels were driven in the Singapore soils by means of the following: many different types of shield (from conventional Greathead to EPBS); the New Austrian tunnelling method (both in soft ground and rock); ancillary measures, including compressed air, jet grouting, chemical grouting and lime piles. This allowed some comparisons to be made between different methods in the same ground, although the variety of conditions encountered limited the number of similar cases. Conclusions that were reached on the relative settlements over tunnels using different methods in similar ground included the following.

(*a*) The NATM in stiff boulder clay resulted in significantly more consistent, and generally less, settlement than occurred over conventional Greathead driven tunnels in the same conditions.

(*b*) Conventional Greathead type shields, with compressed air, gave marginally less settlement in soft marine clay than EPB type shields without compressed air.

(*c*) Jet grouting was highly effective in reducing settlements resulting from tunnelling in soft ground, but the potential side-effect of heave needs to be well controlled to make it a net benefit.

137. It was clear that the amount of settlement was often very dependent on the method of tunnelling used, and the skill with which it was applied. Many of the tunnel drives in Singapore, however, encountered a variety of conditions. It was often the case that the method used for a single drive was a compromise, and not

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necessarily the best (in terms of minimizing settlements), in each of the particular conditions.

138. It many cases, a major portion of the settlements recorded was due to consolidation of the soft clays encountered in or over the tunnels. Further discussion on these settlements is given in response to *Mr Nicholson's* question. However, it is apparent that there is a lot yet to learn about consolidation settlements over tunnels, both in terms of predicting their magnitude and how to minimize them.

139. Comparing the original predictions of settlement and the actual settlements, the largest difference occurred where the tunnel face was in moderately to highly weathered siltstones and mudstones. These materials were originally regarded as reasonably competent, and low settlements were expected when tunnelling through them. In practice, the siltstones and mudstones were often highly fractured, to the point that they sometimes behaved more like gravels than rock, with the added disadvantage of weathering products along the fracture planes acting as a lubricant. It was found that the injection of resin grout into the fractured rocks was effective in stabilizing them, but this was not carried out on a routine basis.

140. In § 57, *Mr Nicholson* asks five questions on marine clay strength. In reply to questions (*a*) and (*b*), the measured shear strength in the overconsolidated crust was higher at Bugis than at Newton Station. This may be because the area around Bugis Station is one of the oldest developed areas of Singapore, while the area of Newton Station has been relatively less developed until recently. A typical Singapore shophouse, of the type found all around the Bugis Station area, would impose an average surcharge of about 30–40 kPa; this alone could account for the difference in crustal strength of up to 10 kPa in measured vane strength.

141. Below the overconsolidated crust, the upper marine clay at Bugis followed a vane strength profile defined by $C_u/\sigma'_v = 0{\cdot}315$,[4] rather than the 0·25 value suggested in the Newton plot. I note that there were very few data below the overconsolidated crust at Newton, compared with the more plentiful data from the thicker marine clay layer at Bugis.

142. In reply to question (*c*) in § 57, there were insufficient oedometer data to make a detailed study of overconsolidation ratios.

143. With reference to question (*d*), a vane strength correction factor of 0·85 was applied to the measured strengths to derive a design value for the diaphragm walls.

144. In reply to question (*e*), no anisotropically consolidated undrained compression or extension tests were carried out for any of the MRT contracts. In the high plasticity marine clay, the use of strengths derived from field vane testing appeared to provide an adequate basis for design.

145. For the slopes shown in Figs 5–7 of Paper 9430, there was considerable discussion on the possible causes of the failures. In all three cases, the major consideration was the appropriateness of the method of analysis and the shear strength values used.

146. For the Orchard Station failure, as indicated by *Mr Nicholson* in § 59, the use of effective stress rather than total stress analysis was found to be more appropriate in the decomposed granite (G4). For those unfamiliar with Singapore, it is worth noting that the term 'G4' is applied to a material which is typically neither strictly granite, nor completely weathered. At Orchard Station, the G4 was predominantly residual soil (weathering grade VI), and was probably derived from

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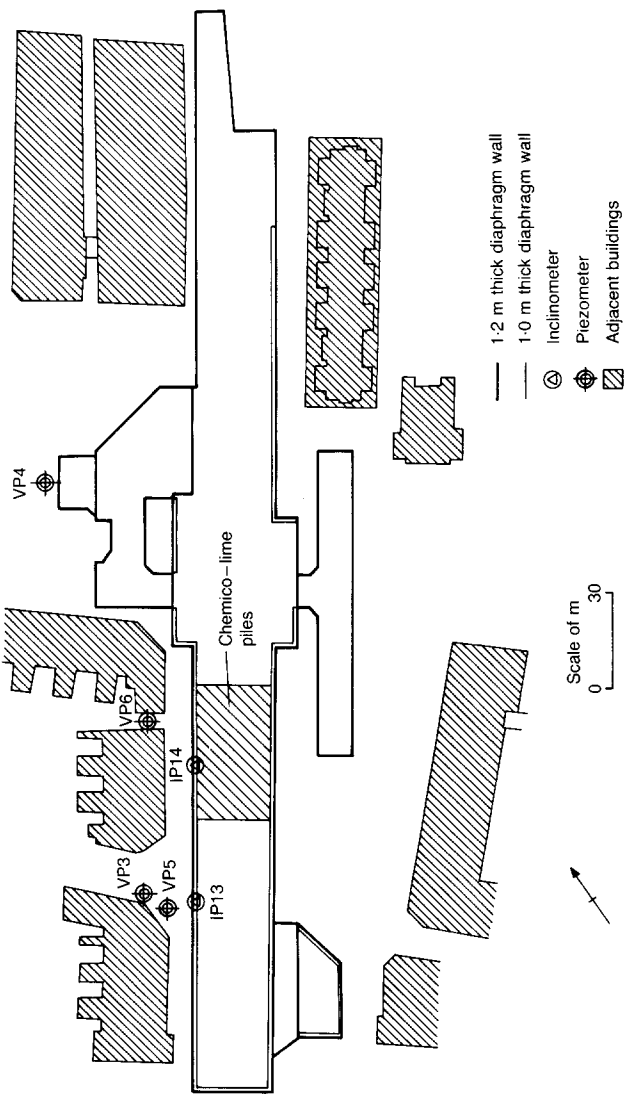


Fig. 2. Plan of Bugis Station

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a granodiorite. The material, on exposure, was similar to a stiff silt. In this soil, the introduction of long, sub-horizontal drains, as shown in Fig. 7 of Paper 9430, was probably of considerable importance in the finally successful completion of these large nailed slopes.

147. City Hall Station, typical boulder clay of the type termed 'S3' was found only close to the base of the excavation. The material forming the sides of the cut was described as 'S4a/b', a soil type found only in conjunction with, and overlying, the true S3. The distinction, geologically, was too fine to make in the necessarily simplified summaries presented in Paper 9430. In practical terms, the S4a/b includes lenses of more granular material, unlike the relatively homogeneous S3. Problems were experienced, as stated by *Mr Nicholson*, with softening immediately under the spreader beams. These beams provided horizontal continuity between rows of anchors, but there was no vertical continuity. The spreader beams could therefore rotate when the soil had softened, leading to a loss of load in the anchors. However, the instrumentation also indicated a deep-seated plane of movement. This movement suggested, on back-analysis, that the material was losing its 'undrained strength' more rapidly than expected. Changes made to the design, to allow this excavation to be successfully completed, included

(*a*) pipe piles to provide vertical continuity between anchors (to address the problem of localized softening at the face of the cut)
(*b*) extra, longer anchors to stabilize the deep-seated failure plane indicated by a 'drained' analysis.

148. At Tanjong Pagar, there may have been an overestimate, in the original design, of the available shear strength in the siltstones and mudstones of the Jurong formation. The rock was fractured, with some weathering along the fracture planes, making it extremely difficult to judge the bulk shear strength.

149. The dewatering at Bugis Station was carried out using only sump pumps. No deep dewatering, using wells or well points, was carried out. There was, typically, a 10–15 m thick 'plug' of very low permeability marine clay below excavation level. The cases at Newton and Bugis Stations were therefore quite different, and it is an interesting question as to why there should have been a similar magnitude of settlement attributable to dewatering at both stations.

150. A plan of Bugis Station is shown in Fig. 2. The location of a few of the monitoring instruments (those whose results will be discussed here) are also shown. Sections, showing the soils and instrumentation in the area of inclinometer IP13, are given in Fig. 3(a) and (b). This area was selected because it included the most complete pore pressure profile, with a piezometer in the Old Alluvium, two in the lower marine clay, and one at the lower/upper marine clay boundary.

151. As the station was excavated, there was a parallel, and dramatic, reduction in pore pressures in the Old Alluvium. Piezometer VP3B, 8·5 m outside the excavation and with a tip 40 m below ground level, showed a 2 m loss of head at only 5 m depth of excavation, and a 5 m loss at 11 m of excavation (see Fig. 4). This led to a concern that underdrainage of the compressible marine clay into the Old Alluvium could be contributing to the significant settlements that were being experienced. It was therefore decided to install a line of recharge wells in this area, where the settlements were largest. These recharge wells, installed fairly close to the piezometers, appeared to have stabilized the situation. However, the settlements continued inexorably, even after the base slab was cast and wall movement had ceased.

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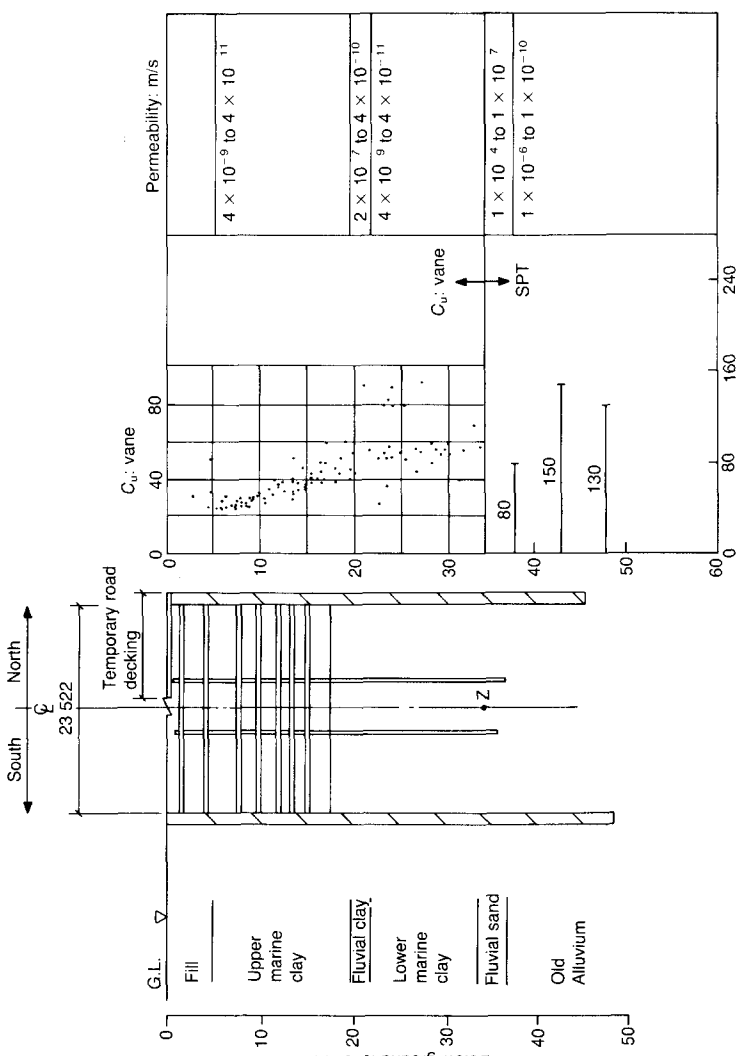
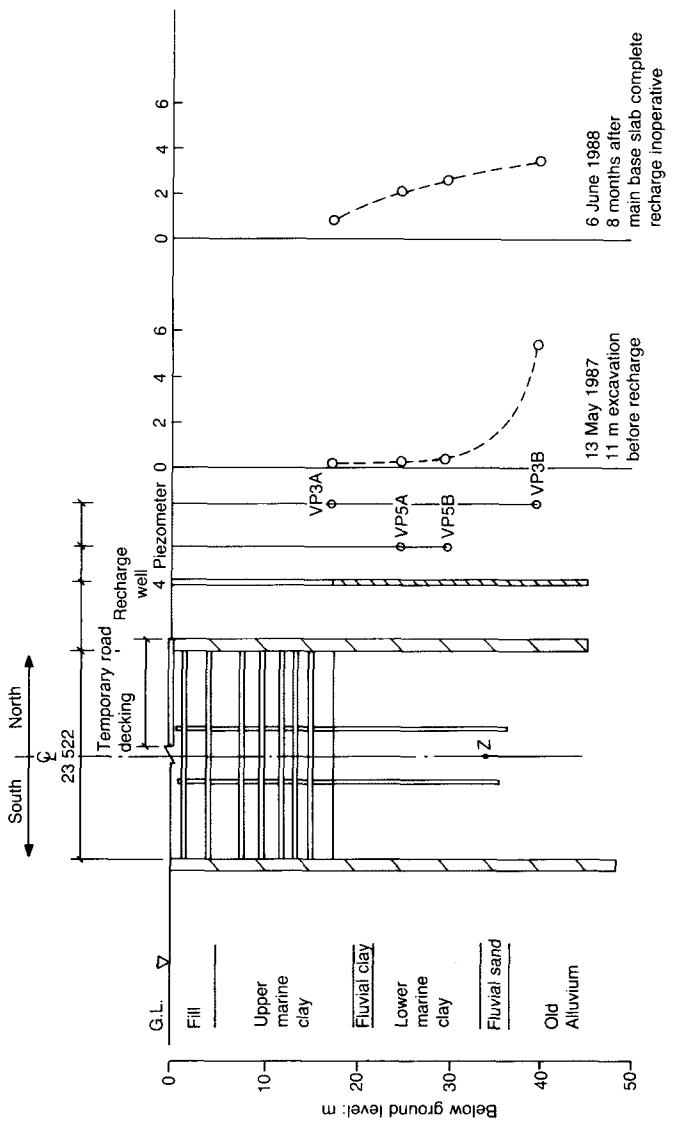


Fig. 3 (above and below). (a) Soils in area of inclinometer IP13; (b) piezometers in area of inclinometer IP13

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(b)

Fig. 3—continued

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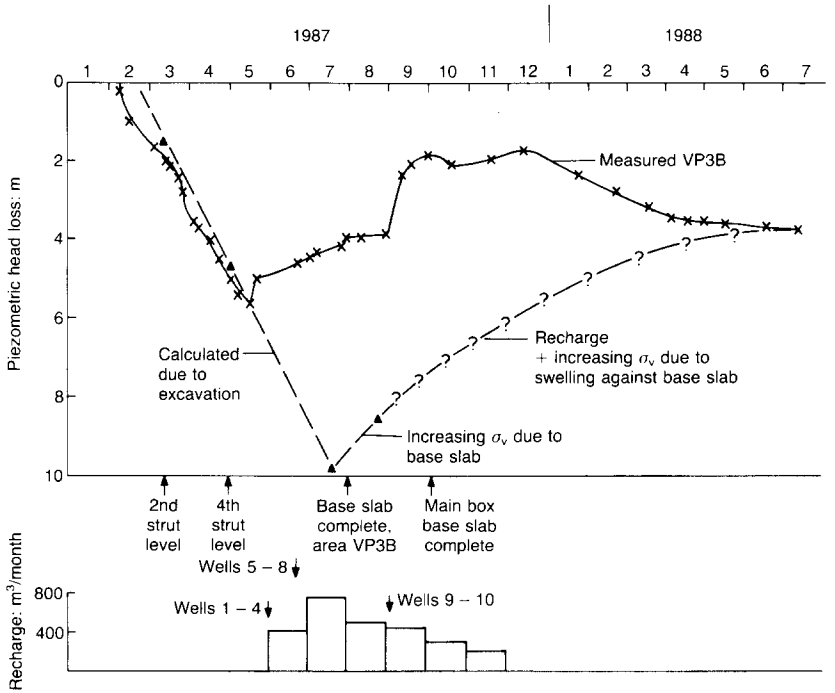


Fig. 4. Pore pressure changes: recharging in area of IP13

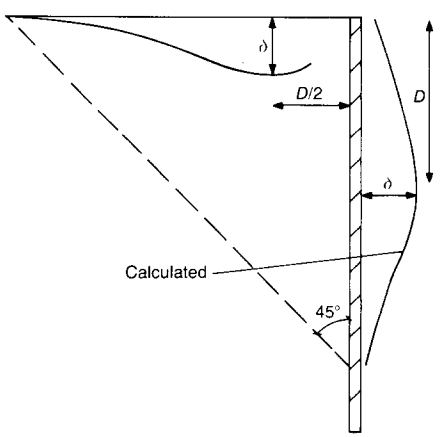


Fig. 5. Relationship between deflexion and settlement in soft clay (after Nicholson[5])

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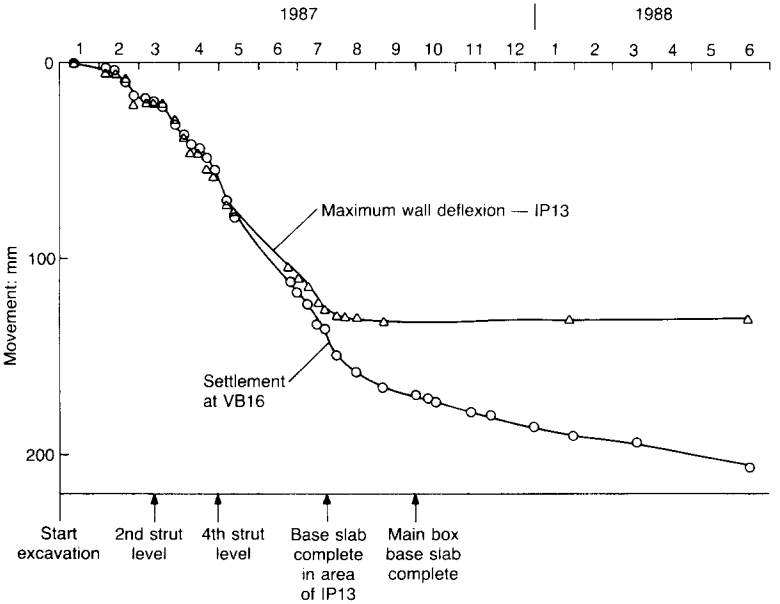


Fig. 6. Comparison between maximum settlement and maximum deflexion—IP13

152. *Mr Nicholson's* proposed relationship between settlement and deflexion (Fig. 5) can be used to estimate the consolidation settlements. In Fig. 6, the settlement at a point 9 m is plotted together with the maximum wall movement (typically 16–20 m below ground level). Mr Nicholson's proposed relationship would suggest that the two values should be approximately equal, excluding consolidation settlements. It can be seen that, down to fourth strut level, the two values did indeed follow a very similar path. After that they diverged, with settlements still continuing 11 months after the base slab in the area of IP13 had been cast. The total consolidation settlement, based on this figure, would have been about 80 mm. The shape of the lateral settlement trough (Fig. 7) also suggests that some 80 mm of consolidation settlement occurred in this area.

153. The recharge wells appear to have treated the symptom (the reduced pressures measured by the piezometers) without being of major benefit in treating the problem. The recharge wells became steadily less efficient with time (Fig. 4), and recharging was eventually discontinued. Fig. 3(b) shows the pore pressure profile recorded in May 1987 (before recharging), and in June 1988 (after the recharging was discontinued). The changes in pore pressure in the marine clay suggest a process of underdrainage into the Old Alluvium.

154. The pore pressure changes due to stress relief at the base of the marine clay (point Z in Fig. 3(b)) can be estimated from

$$\Delta_u = \Delta\sigma_v/2 + \Delta\sigma_h/2$$

The calculated change in pore pressure is shown in Fig. 4. There is good agreement in the calculated values at point Z, and the measured values in VP3B, before the recharging operation. This suggests that

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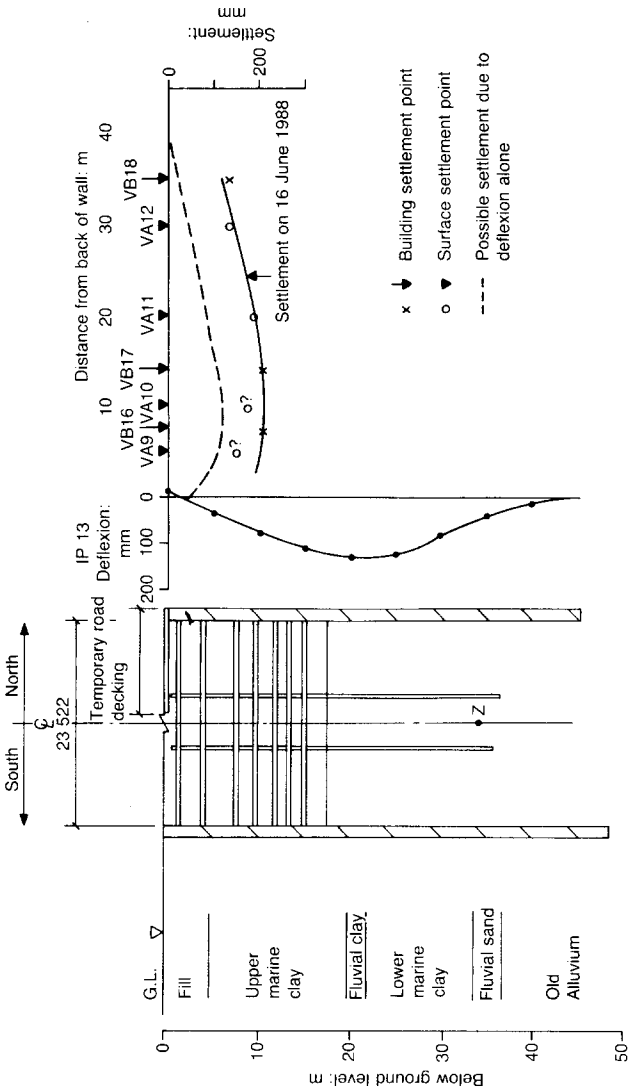


Fig. 7. Deflexion and settlement—IP13

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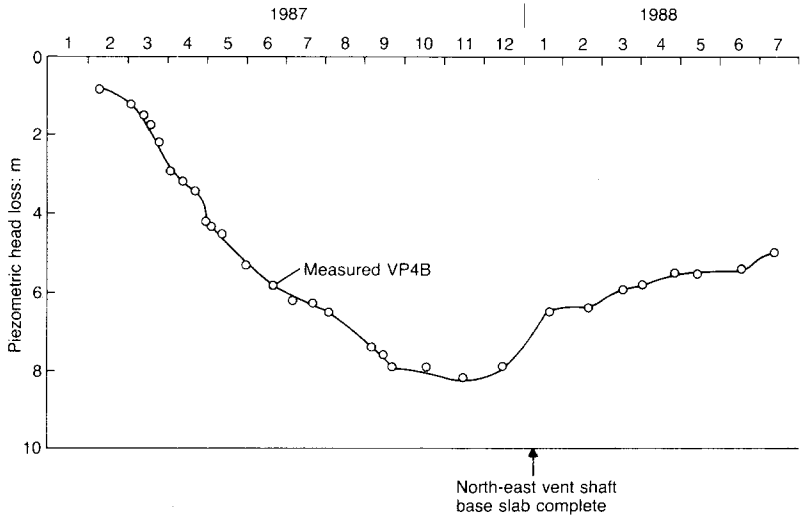


Fig. 8. Head loss in Old Alluvium—VP4B

(*a*) stress relief caused large reductions in pore pressure at the base of the marine clay
(*b*) the Old Alluvium was permeable enough to transmit this pore pressure reduction very rapidly to outside the excavation boundary
(*c*) the Old Alluvium was not permeable enough to provide adequate recharge during the period of the excavation to influence significantly the general reduction in pore pressure
(*d*) had the piezometers not been masked by the recharge wells, a pore pressure reduction of nearly 10 m head would have been measured in the Old Alluvium.

Some confirmation of this last conclusion was obtained from piezometer VP4B, with a tip in the Old Alluvium outside the north-east vent shaft. This piezometer recorded an 8·2 m loss of piezometric head in the Old Alluvium (Fig. 8).

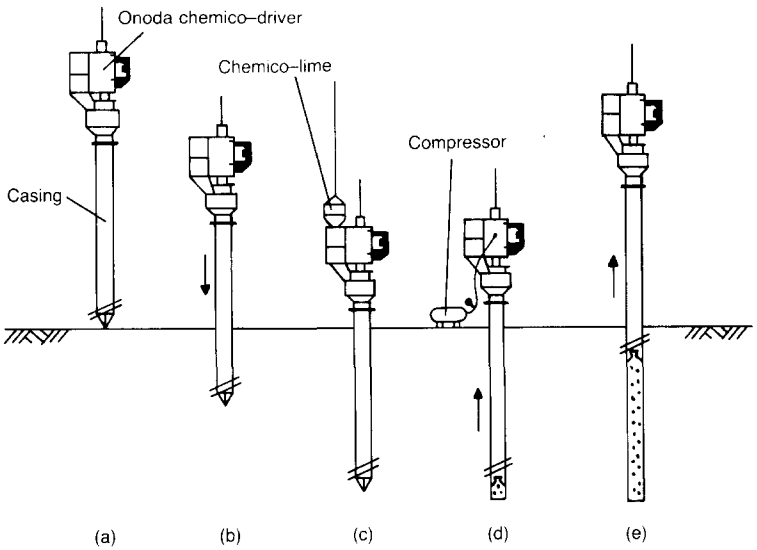
155. From Fig. 3, it appears that the lower marine clay had achieved about 25% consolidation in one year, assuming a 10 m potential loss of head. This would suggest a C_v value of about 7 m^2/year. Choa *et al.* report back-calculated C_v values for marine clay of 1–7 m^2/year at the Changi Airport reclamation. They also report that the higher values were associated with the lower marine clay, which would be consistent with the Bugis measurements.

156. Full details of the Chemico-lime piling are given in Kado *et al.*[6] However, as this paper is not readily available, a brief summary is given below.

157. The lime piles were of the Japanese, rather than of the more familiar Scandanavian type. In the Japanese system, columns of neat lime are formed in the soil; whereas the Scandanavian process is to mix the clay and the lime. The lime in the Japanese system is chemically treated to give the pile considerable strength after hydration.

158. The chemico-lime piles were installed by driving a 0·4 m dia. closed-end mandrel into the ground. Once the mandrel reached the required depth, then the

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(a) The guide leader of the casing is positioned vertically over the determined location.

(b) The casing is screwed down to the required depth at a predetermined rate.

(c) Chemico–lime aggregates, pre-packed in weighted bags, are introduced into the top of the casing through the side hopper.

(d) Compressed air is applied up to the predetermined pressure within the casing.

(e) The direction of rotation is reversed and the casing withdrawn slowly. The shoe of the casing opens and chemico–lime aggregates are discharged, thus forming the pile in the process.

Fig. 9. Installation procedure of chemico-line piles (after Kado et al.[6])

required quantity of chemico-lime was placed inside the mandrel and pressure was applied by means of compressed air (see Fig. 9). When the mandrel was extracted, the shoe at the base on the mandrel opened, so that the chemico-lime remained in the ground.

159. For Bugis Station, two patterns of piling were used, with 1·0 and 1·4 m pile spacings. The effect of the lateral pressure exerted by the swelling of the hydrated columns was to reduce the average water content in the marine clay from 70% to about 55% (1·0 m spacing) and 60% (1·4 m spacing). The changes in the measured vane shear strength are shown in Table 4.

160. The chemico-lime piles were of considerable strength. Samples of the installed piles had an average unconfined compressive strength of 540 kPa. There was some discussion during design as to the benefit that could be obtained from the strength of the piles, as an addition to the increase in the strength of the clay. From the measured wall movements, it is unlikely that any significant benefit was derived from the strength of these discrete, short columns. The benefit obtained

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Table 4. Recorded strength increase following use of chemico-lime piles (after Kado et al.[6])

Chemico pile pitch: m	Depth: m	Average vane shear strength: KN/m²		Original soil vane shear strength: KN/m²	Increase in shear strength: KN/m²	
		Dia.	Side		Dia.	Side
1·0 × 1·0	GL–16·0	58	70	35	23	35
	GL–17·7	70	80	40	30	40
1·4 × 1·4	GL–16·0	55	60	35	20	25

appeared to be mostly from the increase in the strength of the clay. However, there may have been some negative effect from a loss of strength in the upper marine clay as a result of the disturbance during driving of the mandrel.

161. On the question of consolidation settlements over tunnels, four different cases were given in Fig. 23 of Paper 9430. In only one case, namely case (a), did the settlements appear to stop on completion of the permanent lining. This case was not typical of the other MRT tunnels, as it involved the use of a temporary ribs and wooden lagging lining which was far from being water-tight. The other three cases recorded in Fig. 23 involved the use of precast concrete rings as the permanent lining, and so the permanent lining was installed at the tail of the shield. In cases (c) and (d), a water-expansive sealing strip was used, which resulted in extremely dry tunnels (see values for measured flows, given in § 65 of Paper 9430). In spite of this, significant consolidation settlements were still recorded, settlement continuing for over a year after completion of the permanent lining.

162. Hulme *et al.*[7] have suggested that there were possibly three different mechanisms leading to consolidation settlements around the tunnels, each with a quite different surface settlement trough. The three distinct forms of surface trough are shown in Fig. 10; they are thought to show the following.

Form 1. Drainage into tunnel (case (a) of Fig. 23 in Paper 9430).

Form 2. Dissipation of excess pole pressures caused by tunnelling at a pressure less than the in-situ earth pressure (cases (b) and (c) of Fig. 23).

Form 3. Dissipation of excess pore pressures caused by tunnelling with a pressure greater than the in-situ earth pressure (case (d) of Fig. 23).

There was very little measurement of pore pressures to confirm these suggestions. However, it is of interest that for both forms 2 and 3 settlements, there was a distinct delay of 3–4 weeks in the onset of the consolidation settlements. This delay was recorded only in those tunnels where a water expansive seal was used on the segmental lining.This delay does suggest that seepage into the tunnel was not the underlying cause of this settlement.

163. For the form 3 consolidation settlements, the delay is clear in Fig. 23 of Paper 9430. This is not so for the compressed air tunnel (case (c)) because of the effects of using a logarithmic time-scale. If the progress of the consolidation settlements follows an exponential curve, then this curve becomes a staight line when

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Example 1	*Example 2*	*Example 3*
Single tunnel 6·41 m OD drum digger	Twin tunnels 16 m c/c 5·91 m OD Greathead shields	Twin tunnels 16 m c/c 5·93 m OD earth pressure balance
Depth to axis: 13 m	Depth to axis: 15·3 m (E/B) 16·0 m (W/B)	Depth to axis: 18·9 m
Air pressure: 0·9 bars	Air pressure: 1·3 bars (E/B) 1·25 bars (W/B)	Face pressure (shove): 3·5 bars (E/B) 3·8 bars (W/B) (rest) : 2·5 bars (E/B) 2·3 bars (W/B)
Temporary lining of ribs and wooden lagging Permanent lining of in-situ concrete placed after tunnel depressurized	Precast concrete lining with hydro-swelling sealing strip	Precast concrete lining with hydro-swelling sealing strip

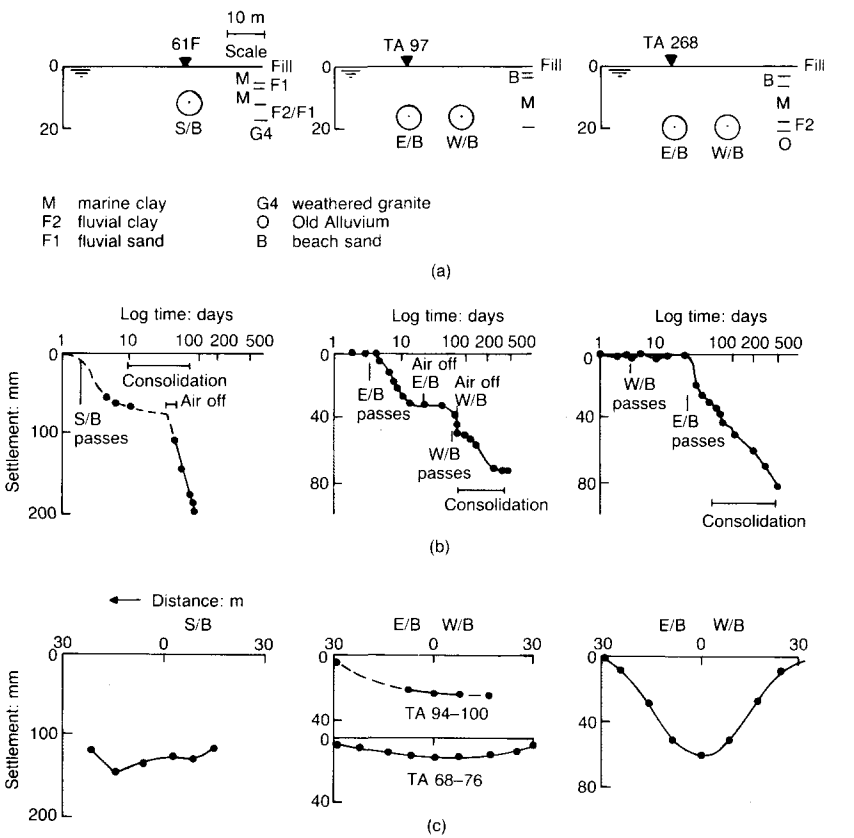


Fig. 10. Three different patterns of consolidation settlements: (a) tunnel details; (b) settlement/log. time for points indicated in (a); (c) lateral settlement trough for periods indicated in (b) (after Hulme et al.[7])

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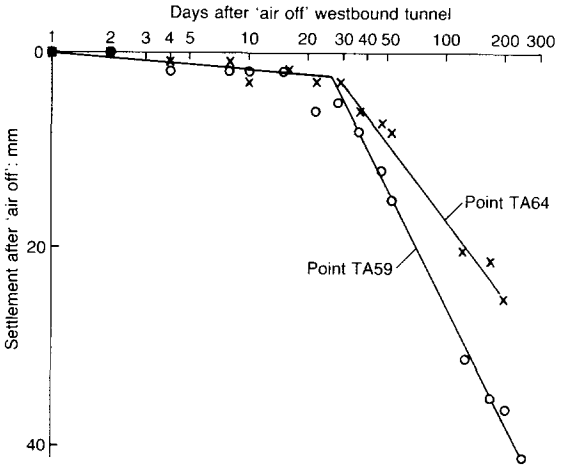


Fig. 11. Development of settlements over tunnel centre-line following depressurization of tunnel

the time-scale is logarithmic, but only if the start of the curve is at the origin. If the start of the consolidation process is not at the origin, then the progress of the consolidation settlements starts to curve. For the form 3 settlements (case (d)), the start is reasonably close to the origin, and the distortion is not significant. However, for case (c), the start of the consolidation process, with the depressurization of the Westbound tunnel, is far from the origin. This distortion, combined with the compression of the data points due to the scale, has the result that the progress of the consolidation settlements is unclear in case (c) of Fig. 23.

164. A clearer picture of the consolidation settlement is obtained if the settlement after 'air off' only is plotted. Fig. 11 shows the settlement/log. time plot for this period at two locations. These show a clear delay, of about four weeks, in the onset of the consolidation settlement. To demonstrate that this was a consistent trend, the settlements recorded at 11 different points, covering some 300 m of twin tunnel, are summarized in Fig. 12. In this figure, the consolidation settlement at each point is normalized by dividing it by the consolidation settlement recorded at the same point 202 days after 'air off'. Day 202 was chosen because some of the points were lost after this date. It can be seen that the trend established in Fig. 12 was a consistent one.

165. Also plotted in Fig. 12 are the pore pressures recorded in a pore pressure probe jacked out from the Westbound tunnel into the pillar of clay between it and the Eastbound tunnel. This piezometer showed that a significant excess pressure of nearly 40 kPa built up over the five weeks after 'air off', and that the start of the dissipation of those pressures and the onset of the more rapid consolidation settlements were reasonably coincident.

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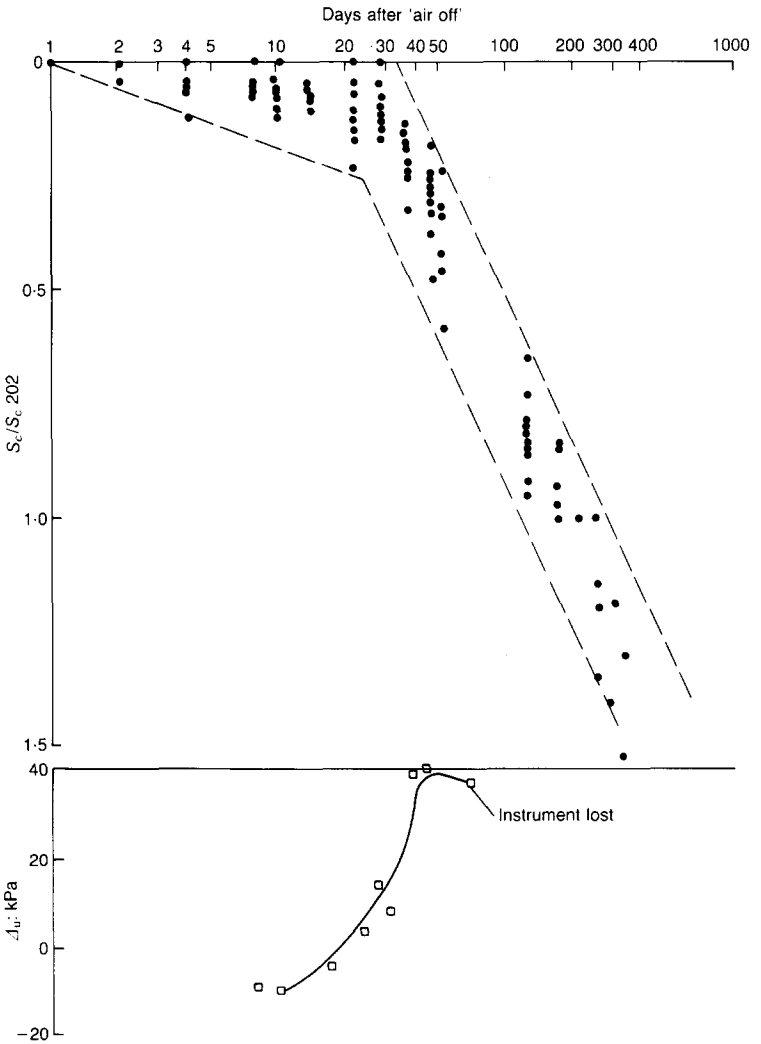


Fig. 12. Development of consolidation settlements and recorded pore pressure changes (S_c is consolidation settlement; S_c 202 is consolidation settlement, day 202)

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Corrigenda

The correct version of Table 3 of Paper 9428 is shown below.

Table 3. Contingency allowances

Work categories	%
Elevated sections between stations	5
Elevated stations	5
At grade construction	5
Transitions	10
Underground cut and cover between stations	10
Underground stations	10
Driven tunnel	15
Trackwork	2·5

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Corrigenda—continued

The correct version of Table 2 of Paper 9430 is shown below.

Table 2. Temporary work for underground stations

Station	Maximum depth of excavation: m	Typical soil sequence*	Retaining system used	Special measures
Braddell	14·9	1F, G4	0·6 and 0·8 m diaphragm walls	
Toa Payoh	13·5	4F, 4K, G	Sheet piles	
Novena	14·7	½F, 14·2K, G	Sheet piles	
Newton	14·3	3F, 13K, G	0·8 m diaphragm walls	Jet grouting
Orchard	21	½F, G	Nailed slopes	
Somerset	16·2	2F, 8K, G	0·6 m diaphragm walls or sheet piles	
Dhoby Ghaut	16·1	1F, 10K, S	Sheet piles	
City Hall	22·3	3F, 2K, S3	8 m deep sheet piles over slopes, anchors	
Raffles Place	25·2	2F, S3	King piles and shotcrete lagging	
Tanjong Pagar	17·9	½F, S	Slopes, anchors	
Outram	13·9	2F, 3K, 3	8 m deep sheet piles over king piles and timber laggings	
Tiong Bahru	14·1	1F, S	King piles and shotcrete lagging	
Bugis	18·3	1F, 34K, O	1·2/1·0 m diaphragm walls	Lime piles
Lavender	16·5	3F, 20K, O	1·0 m/diaphragm walls	
Marina Bay	16·4	12F, 24K, O	Composite H pile/sheet pile	Underwater excavation

*F = fill
G = Granite, S3 = Jurong } including weathered rock
K = Kallang
S = Boulder Bed
O = Old Alluvium

Example: 3F, 13K, G = 3 m of fill overlying 13 m of Kallang deposit overlying granite (in this case completely weathered granite).